



Case Report

Unexpected death due to spontaneous splenic rupture: A rare case in splenic angiosarcoma



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ABSTRACT

Splenic angiosarcoma (SA) is a rare disease that can cause spontaneous splenic rupture leading to unexpected death. The rare incidence and non-specific clinical presentations made the early correct diagnosis of SA impossible in clinical practice. Even with medical intervention, 80% of patients died within 6 months after diagnosis. Here, we report a man who had complained of abdominal distension for 2 weeks and succumbed to the disease nine hours after admission. Diagnosis of hepatic and splenic angiosarcomas was based on post-mortem histopathological examination and immunohistochemistry, and the cause of death was hemorrhagic shock caused by spontaneous splenic rupture secondary to SA. This present case was the first forensic autopsy of spontaneous splenic rupture secondary to SA, which can highlight the diagnosis of rare diseases in forensic practice, and forensic pathologists should bear in mind these rare diseases even in routine practice.

1. Introduction

The characteristics and location of spleen making it the most fragile organ in abdominal cavity, it is a common view that the splenic rupture usually combined with blunt abdominal trauma. In forensic practice, the real cause underlying the splenic rupture should be thoroughly investigated, and spontaneous splenic rupture should exclude minor trauma [1]. Non-traumatic splenic rupture always attributed to pathological causes and splenic neoplasm accounting for almost 1/3 of non-traumatic splenic rupture, ranking first of all causes which including infectious, inflammatory disease, medication and medical treatment and so on [2,3].

Angiosarcoma is a malignant tumor arising from vascular endothelial cells. The primary focuses of this disease often locate at breast, skin and soft tissues, and this malignancy rarely occurs in the spleen with an incidence range from 0.15 to 0.26/1000000 [4,5]. Splenic angiosarcomas was first reported by Langerhans in 1879, and since then about 200 cases were reported worldwide [5,6]. Primary splenic angiosarcoma (PSA), although rare, is the most common spleen malignancy. PSA usually spreads to the liver through hematogenous at early stage of the disease, in a Chinese literature, liver was the most common second organ with an incidence as high as 78.2% [7]. Hepatic angiosarcoma is a rare aggressive tumor. Worldwide, only about 200 cases are diagnosed each year, but this is the most common primary hepatic stromal tumor in adults, accounting for 2% of all primary liver

malignant tumors. Compared to the other sites of angiosarcoma, hepatic and splenic angiosarcomas is very aggressive, and most patients are found at an advanced stage, and therefore presents a poor prognosis [8,9]. Almost all patients died within 6 months after diagnosis irrespective of treatment and some were diagnosed after death [8]. We present a case of spontaneous rupture of SA and diagnosed posthumously.

2. Case summary

A 44-year-old male in coma was hurried taken to the hospital due to abdominal distension for 2 weeks. He was an accountant in a private company without any known risk factors for hepatic or splenic angiosarcoma. Physical examination revealed no significant abnormality in appearance. The laboratory examinations revealed: red blood cell count $2.68 \times 10^{12}/L$, hemoglobin 81.0 g/L, platelet count $51.0 \times 10^9/L$, γ -glutamyl transpeptidase 158 U/L, pro-thrombin time>35 s, activated partial thromboplastin time>100 s, fibrinogen < 1 g/L, D-dimer 32.93 mg/L. alpha-fetoprotein(AFP), carcinoembryonic antigen(CEA), carbohydrate antigen 125(CA125), carbohydrate antigen 153(CA153), and carbohydrate antigen 19-9(CA19-9) revealed normal. Serology for hepatitis A, B and E tested negative. A computed tomography (CT) scan of the abdomen was performed and revealed hepatomegaly, splenomegaly and diffuse low signal changes of the liver and spleen parenchyma. On admission, 100 ml 0.9% NaCl, 250 ml amino acid reagent,

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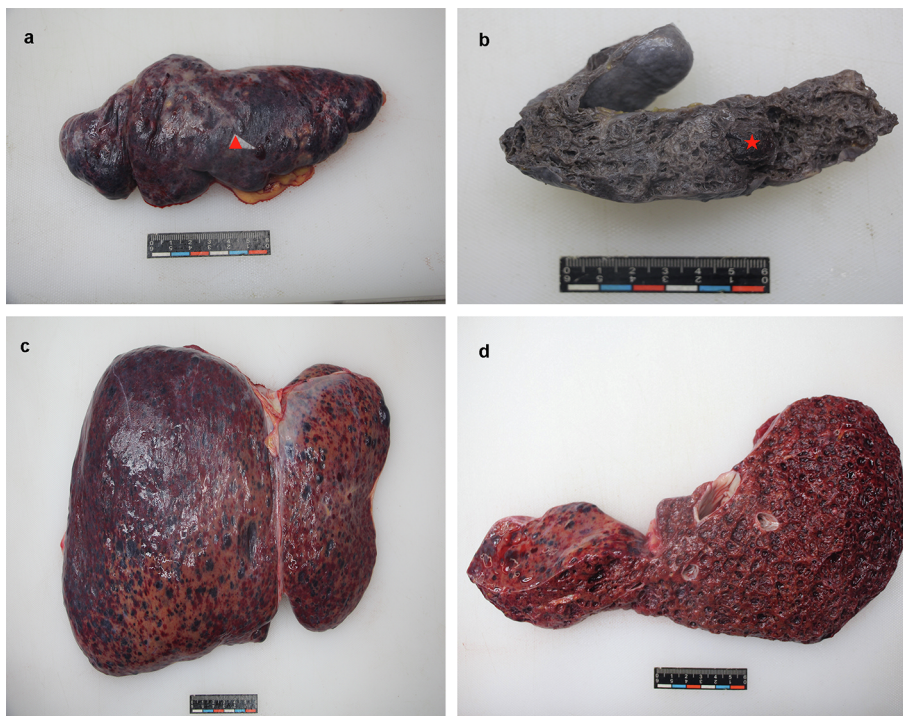


Fig. 1. Gross findings of spleen and liver. Tumor diffusely infiltrated the liver and spleen. a A 0.5 cm curved rupture was seen on the diaphragmatic surface of the spleen (arrows) and (b) section of the spleen demonstrated diffuse honeycomb or cystic space like liver with different sizes; Surface of liver was uneven and heterochromatic and (d) section of the liver showed diffuse honeycomb or cystic space with different sizes like spleen. ▲ rupture site of spleen, ★the hematoma in the spleen under the rupture site.

250 ml 5% glucose saline infusion were arranged and the patient was awake, whereas, eight hours later, the patient suddenly lost consciousness with respiratory and cardiac arrest, and finally died one hour later.

A medico-legal autopsy was performed 10 h post-mortem as his direct relatives suspect the existence of medical malpractice. External examination revealed no significant injuries except for injection marks. The faint lividity, pale extremities, pallor cornea and empty of cardiac chamber indicated systemic anemia.

Upon open the abdominal cavity, it was filled with hemorrhage. To investigate the original site of intro-abdominal hematoma, we cleared all the hemorrhage and the quantity of bleeding was about 3500 ml. A 0.5 cm curved rupture was seen on the diaphragmatic surface of the spleen (Fig. 1a), on the section of the spleen, a hematoma sizing 6.0 cm × 5.0 cm × 0.5 cm under the rupture.

The liver weighed 2100 g and the spleen 220 g. Both organ surfaces were uneven and heterochromatic (Fig. 1a, c). In the section of the liver and spleen, diffuse honeycomb or cystic space with different sizes were seen (Fig. 1b, d).

Pathologic examination showed vascular structures of different sizes and anastomosing with each other combined with various hematomas near the rupture. To further evaluate the time of splenic rupture, Masson and Prussian blue staining were performed on the rupture site of spleen, with positive results (Fig. 2).

Upon pathologic examination of the liver and spleen, vascular structures of different sizes and anastomosing with each other were notable. Neoplastic cells were arranged irregularly in nodular, fascicular or nest pattern, and some were papillary and protruded into the vascular lumen or cystic spaces. The vascular spaces were variably capillary-like, slit-like, cavernous, honeycomb-like to dilated sinusoidal (Fig. 3a, g).

Neoplastic cells were variably round, spindly to irregular, and similarly to the nuclei. Vesicular nuclear chromatin and mitotic figures could be seen. We screened a series of biomarkers by using immunohistochemistry to make a further diagnosis. Vascular differentiation markers CD31, CD34, FVIIIIRag, transcription factor ERG, Fli-1, histiocytic differentiation markers CD68, Vimentin, EMA, Desmin, and S100 were finally selected for the correct diagnosis. The tissues of the

liver and spleen were stained positively by CD31, CD34, FVIIIIRag, ERG and Vimentin (Fig. 3b–f, i–l), and negatively by EMA, Fli-1, Desmin, CD68 and S100.

According to the pathologic findings along with a detailed review of medical history, we concluded that the deceased was an angiosarcoma patient and the cause of death was hemorrhagic shock caused by spontaneous splenic rupture second to splenic angiosarcoma.

3. Discussion

Angiosarcoma is a rare, heterogeneous sarcoma subtype, accounting for less than 1% of all sarcomas [4]. While splenic angiosarcoma occurs primarily in individuals that above 40 years old, hepatic angiosarcoma tend to affect these individuals older than 60, and a slight male predominance was observed in both types [4,10]. Although the environmental carcinogens (thorium dioxide, vinyl chloride monomer, inorganic arsenic and use of anabolic steroids) and benign hemangioma have been claimed to be related agents, the real relationship were not clearly illustrated, and the etiology remains uncertain in most cases [10]. In present case, the related investigation showed negative any known risk factors, so the real cause is unknown.

The clinical presentation of angiosarcomas varied according to primary site or histological subtype. Compared to the skin and breast angiosarcomas, splenic angiosarcoma and hepatic angiosarcoma are more hidden. The initial symptoms of splenic angiosarcoma and hepatic angiosarcoma are nonspecific, including abdominal pain, abdominal distension, fatigue, anorexia and weight loss [9]. We also searched “spontaneous splenic rupture angiosarcoma” in pubmed during 2004–2019, and ruled out the only abstract and non-English articles, and finally we carefully reviewed 9 previous spontaneous splenic rupture due to splenic angiosarcomas, splenomegaly was common, whereas in present case, the rather normal spleen maybe the reasons fail to make the correct diagnosis (Table 1) [11–19].

Although imaging examination is important to the early diagnosis of hepatic and splenic angiosarcomas [20,21], a definitive diagnosis of angiosarcomas can't be obtained because of its various morphological patterns [22]. Morphologically, hepatic and splenic angiosarcomas manifests as solitary, dominant mass with satellite nodules, multiple

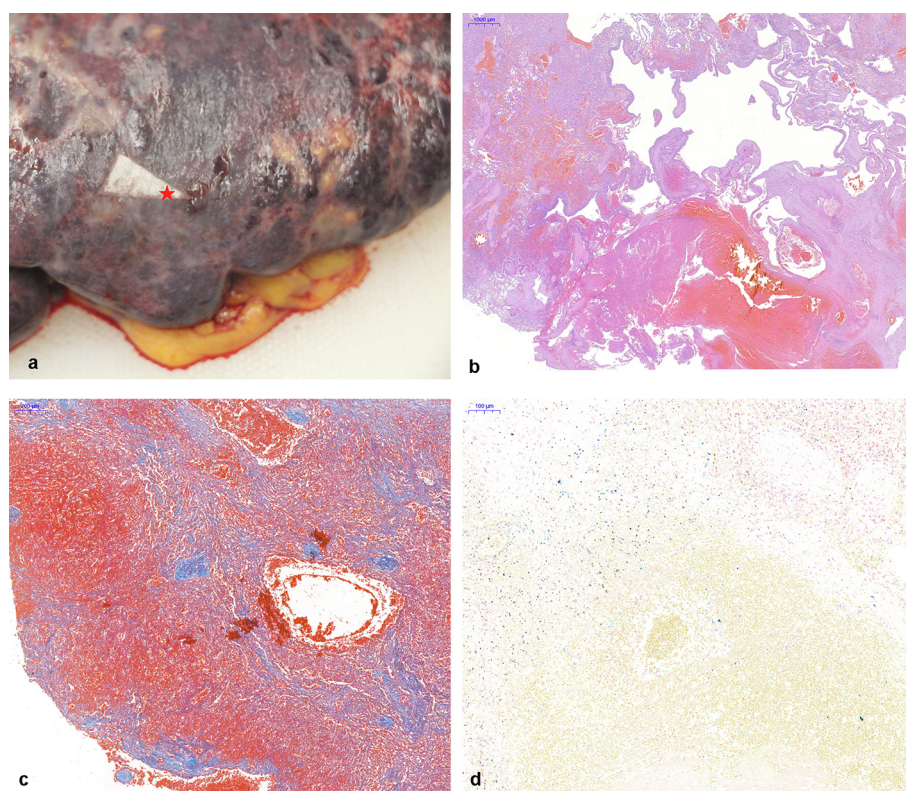


Fig. 2. The rupture of spleen and histopathological examination. a, ★the curve rupture of spleen; b, HE 10 $\times$; c, Masson 50 $\times$; Prussian blue, 100 $\times$.

nodules and diffusely infiltrating lesion [23]. In our case, the imaging of CT scan which showed splenomegaly and diffuse low signal changes of the liver and spleen parenchyma may indicated the presence of hepatic and splenic angiosarcomas.

Since the patients of PSA present no significant history of exposure to risk factors and non-specific clinical presentations, tumor marker or imaging finding, it is impossible to make a correct diagnosis at an early stage [24]. The diagnostic value of microscopic examination is reasonably emphasized. Microscopic examination is the most reliable method to ascertain diagnosis of angiosarcomas [25]. Histologically, angiosarcoma is characterized by spindle-shaped cells, which form a variety of vascular channels, such as dilated sinusoidal, cavernous spaces and slitlike freely anastomosing vessels, and need make differential diagnosis with other tumors [26].

Although splenic angiosarcoma is rare, vascular tumors are the most common non-hematopoietic tumors that primary arise from spleen. The pathological changes in spleen and liver remind us the possibility of angiosarcoma, and need to distinguish from epithelioid hemangioendothelioma, Kaposi sarcoma, fibrosarcoma and so on. Previous studies showed vascular differentiation markers, transcription factor ERG is critical to make the correct diagnose [25,27,28]. Immunohistological examination positive of vascular differentiation markers CD31, CD34, FVIIIIRag, transcription factor ERG and Vimentin combined with medical history help us distinguish present case from other diseases.

Splenic and hepatic angiosarcomas are highly malignant vascular tumors, it is a common view that liver involved ranks first in splenic angiosarcoma and hepatic angiosarcoma also can affect spleen. Various malignancy splenic tumors usually found during autopsy. In a consecutive study of 1280 splenectomy specimens, metastases were found in 14 cases, among which 2 were metastatic sarcoma [29]. Previous studies indicate splenic metastatic tumors can cause splenomegaly or spleen rupture [30]. The clinical and pathological characteristics of splenic and hepatic angiosarcomas made the primary affected organ in present case unillustrated.

Splenic bleeding is common in forensic practice, and when referred to splenic bleeding, splenic vessel rupture maybe overlooked. It is a common view that splenic aneurysm ranks first of all visceral aneurysms, and any diseases that can alter the structure of spleen or splenomegaly may make spleen more fragile. Trauma ranks first in splenic rupture. In forensic practice, thoroughly autopsy investigation and pathological examination should be performed in splenic bleeding cases to exclude the underlying vessel rupture and pathological process even in traumatic rupture [1]. In present case, without any known trauma history in case investigation, the normal external examination, the medical history and pathological examination strongly indicated spontaneous splenic rupture.

Splenic and hepatic angiosarcomas are highly malignant vascular tumors with a poor prognosis and may cause splenic rupture. Forensic pathologists should bear in mind this rare disease and make thoroughly investigation to make correct diagnosis.

4. Compliance with ethical standards

Ethical approval

This study was approved by the Ethics Committee of Tongji Medical College, Huazhong University of Science and Technology, Wuhan, Hubei, China.

Informed consent

Written informed consent was obtained from the family members of the patient.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to

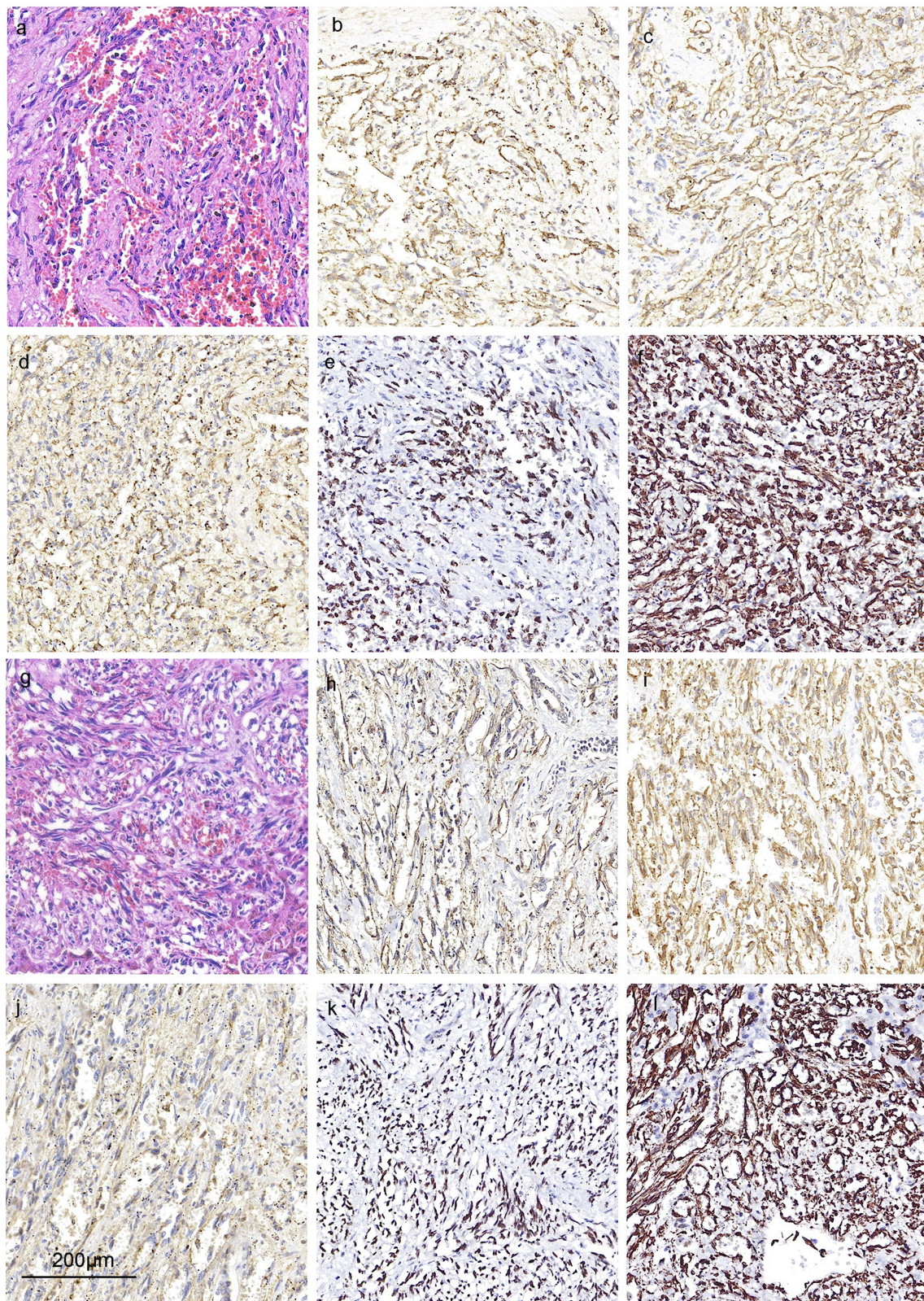


Fig. 3. Pathology features of splenic and hepatic angiosarcomas. a, g The vascular spaces demonstrated capillary-like or slit-like, HE, 200 $\times$. b–f, i–l Immunohistochemistry magnification view ($\times 200$) showing the tumor cells were positive for CD31, CD34, FVIIIIRag, ERG and Vim in spleen and liver respectively.

Table 1
Detailed information of previous case reports about spontaneous splenic rupture caused by splenic angiosarcoma.

Author/publication year	Age (years old)/sex	Onset symptom	Metastatic	Diagnosis after	Spleen	Immunohistochemical staining	Outcome
Cao et al. 2019 [11]	47/female	Unknown	Liver	Splenectomy	unknown	CD31	Died 51 month later
Özcan et al. 2018 [12]	65/female	Abdominal pain	Liver, bone	Splenectomy	splenic mass	CD31	unknown
Sara et al. 2017 [13]	49/male	Diffuse abdominal pain		Splenectomy	enlarged spleen	CD34, factor VIII, CD31 CD68pgm1	Died within 1 year
Duan et al. 2013 [14]	65/male	Diffuse abdominal pain and distension		Splenectomy	1650 g	Vimentin, CD31, CD34, factor VIII, a1ACT and SMA	Unknown
Xu et al. 2015 [15]	77/female	Diffuse abdominal pain and distension	lung	Splenectomy	1550 g	CD31, CD34,	Died 2 weeks post splenectomy
Badiani et al. 2013 [16]	30/male	abdominal pain		Splenectomy	990 g	CD31	Died
Alexandrino et al. 2013 [17]	80/female	Syncope, hypotension and vomiting		Splenectomy	242 g	CD31, CD34,	6 months still well
Maier et al. 2004 [18]	37/female	Abdominal pain		Splenectomy	520 g	CD31, CD34, factor VIII,	Died within 6 months
Tataria et al. 2005 [19]	27/female	Abdominal distension, syncope		Splenectomy			Died within 2 months

influence the work reported in this paper.

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